Computer Networks 149 (2019) 29–42

Contents lists available at ScienceDirect

Computer Networks

journal homepage: www.elsevier.com/locate/comnet

A lightweight anonymous user authentication and key establishment scheme for wearable devices

Ankur Gupta[a,*], Meenakshi Tripathi[a], Tabish Jamil Shaikh[b], Aakar Sharma[b]

[a] Department of Computer Science and Engineering, Malaviya National Institute of Technology, Jaipur, Rajasthan, India
[b] Department of Computer Science and Engineering, Indian Institute of Technology Jammu, Jammu and Kashmir, India

ARTICLE INFO

Article history:
Received 24 June 2018
Revised 20 October 2018
Accepted 16 November 2018
Available online 17 November 2018

Keywords:
Internet of things
Wearable devices
Key establishment
Mutual authentication
BAN-Logic
AVISPA

ABSTRACT

The paradigm of the Internet of Things (IoT) is a system of interconnected objects that can be accessed globally with the help of the Internet. IoT has numerous applications including healthcare wherein the wearable devices can sense the data from the body of the patient and sent it to the concerned doctor for remote monitoring. It is essential that these devices should operate in real time and generate the precise data as life-critical decisions are made based upon the data received from these devices. The infrastructure of IoT is very heterogeneous and dynamic therefore vulnerable to the threat of security and privacy. One of the most significant challenges in IoT is the authentication of the devices before sending the data so that we can have confidence in the received data. As IoT devices are resource constraint, there is a need for lightweight authentication scheme for them. In this paper, we introduce a new authentication technique for communication on the Internet of Things (IoT) using simple XOR and one-way Cryptographic hash function. The proposed protocol not only provides security but also maintains equilibrium between the efficiency and communication cost. The security of the proposed protocol is evaluated using BAN-Logic, the widely accepted AVISPA tool and the informal security analysis. Then, the comparison of the proposed protocol with the existing related schemes is shown concerning security features, communication, and computational cost.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

The Internet of Things (IoT) plays a vital role in the healthcare industry [1]. The progressions in innovation are empowering the production of small computers more than ever. The wearable technology enhances the IoT applications to upgrade the quality of human life [2,3]. The Wearable devices like a smartwatch, smart glasses, fitness tracker etc. [4] are used to collect real-time user's health data like step-count, sleep-cycle, heart-rate and pulse-count. Medical applications typically consist of low power, low cost and lightweight devices [5]. In IoT based healthcare services, the information collected by the wearable devices are sent to the mobile device containing sensitive and confidential data that must be accessed only by the authorized personnel. Authentication is the primitive step for any secure system, it is the process of verifying a claimed identity. A wearable device must authenticate the user before providing confidential information. The wearable devices are restricted for capabilities regarding communication and processing power. Therefore, lightweight mutual authentication and key establishment techniques are essential in IoT to resist several known attacks like man-in-the-middle, impersonation, replay, and stolen-verifier attacks.

The authentication model uses three types of cryptosystems to achieve the mutual authentication, user anonymity, privacy-preserving, and perfect forward secrecy, namely: symmetric cryptosystem, asymmetric cryptosystem, and hybrid cryptosystem [6]. Symmetric-cryptosystem based protocols need less computation power, however, at a cost of high memory consumption to save all the symmetric keys required in a network. Still, the efficiency of the symmetric-cryptosystem based protocols succeeds the asymmetric-cryptosystem based techniques.

In this paper, we propose a new lightweight anonymous user authentication scheme for the communication on the Internet of Things (IoT) based only on simple XOR and one-way cryptographic hash function. The proposed protocol not only provides security but also very efficient regarding computation cost as it is based on symmetric cryptographic technique.

* Corresponding author.
E-mail addresses: 2015rcp9543@mnit.ac.in (A. Gupta), mtripathi.cse@mnit.ac.in (M. Tripathi).

https://doi.org/10.1016/j.comnet.2018.11.021
1389-1286/© 2018 Elsevier B.V. All rights reserved.

1.1. Our contribution

The following are the main contributions of this paper:

- A light weight authentication model for the wearable sensing devices is presented in the paper.
- The proposed protocol uses only simple XOR and one-way cryptographic hash function to share the session key between the mobile terminal and a wearable sensing device.
- The proof of establishing a secure session-key is provided using BAN-Logic.
- AVISPA simulation tool is used for the formal security verification to show the resilience to various known attacks under Dolev–Yao attack model.
- The security against the various well-known attacks have also been provided using the informal security analysis.
- Finally, the efficiency of the proposed protocol is shown in terms of computational and communication cost.

1.2. Organization of the paper

The remainder of the paper is organized as follow. Section 2 discusses the essential mathematical preliminaries that are required to design and evaluate the proposed protocol. Section 3 discusses the overview of some existing related work in this field. Section 4 discusses about the system model used in the proposed scheme. Section 5 presents the proposed scheme in details. Section 6 provides the security analysis of the proposed protocol. Section 7 gives the comparative analysis of the proposed protocol regarding communication and computation cost with related existing schemes. Finally, Section 8 concludes the paper.

2. Mathematical preliminaries

This section quickly discusses the fundamental concepts of the Bitwise XOR operation and the one-way cryptographic hash function which we have utilized in our protocol for security purpose in the wearable devices.

2.1. One-way cryptographic hash function

A one-way cryptographic hash function takes a string $S \in \{0, 1\}^*$ of arbitrary-length as an input and produces a fixed-length string (say, n-bits) $Y \in \{0, 1\}^n$ as an output, called the hash value or a message digest. The symbolic representation of hash function can be seen as $h(\cdot)$: $S \rightarrow Y$. Following are the main properties of the cryptographic hash function -

- *Easiness:* For a given input string $S \in \{0, 1\}^*$, $Y = h(S)$ is easily computable where $Y \in \{0, 1\}^n$.
- *Pre-image resistance:* From a given message digest Y, it is computationally infeasible to find a string $S \in \{0, 1\}^*$, where $Y = h(S) \in \{0, 1\}^n$.
- *Weak-collision resistance:* It is computationally hard to find any other input $S' \in \{0, 1\}^*$, with $S \neq S'$ such that $h(S) = h(S')$.
- *Strong-collision resistance:* It is computationally infeasible to find a pair of inputs $(S, S') \in \{0, 1\}^*$, with $S \neq S'$, such that $h(S) = h(S')$.

Some of the popular cryptographic hash functions are MD5 [7], SHA-1 [8], RIPEMD160 [9], BLAKE2 [10] and SHA-3 [11].

2.2. Bitwise XOR operation

A bitwise XOR operation takes two equal length (say, n-bits) Bit pattern A, $B \in \{0, 1\}^n$ as input and executes the logical Exclusive-OR operation on the corresponding pair of bits. The output of the resultant operation is 1 if both the bits are different otherwise 0. The main properties of bitwise XOR operation are -

- For a given input A, $B \in \{0, 1\}^n$, it is computationally fast to calculate the output $Z = A \oplus B$, where $Z \in \{0, 1\}^n$.
- For a given output $Z \in \{0, 1\}^n$ and any one input (say, $A \in \{0, 1\}^n$), it is fast and reversible to calculate the other input $B \in \{0, 1\}^n$, i.e. $B = Z \oplus A$.
- For a given output $Z \in \{0, 1\}^n$, it is computationally infeasible to find both the inputs A, $B \in \{0, 1\}^n$, such that $Z = A \oplus B$.

3. Related work

The first lightweight user authentication protocol for resource-constrained devices based on XOR operation and hash function only was presented by Wong et al. [12] in 2006. However, their symmetric protocol was susceptible to various attacks such as replay, forgery and stolen-verifier attack as shown by Tseng et al. [13], thus it was inappropriate for use. In 2009, Das [14] also found these attacks and the same login-id threat in Wang et al.'s scheme and thus presented the improved protocol by introducing password-based third-party user authentication using gateway node. However, this smart-card based scheme failed to achieve mutual authentication and key agreement, and was also susceptible to the insider attack, node-capture attack, DoS attack, and impersonation attack. Khan and Alghathbar [15] proposed a scheme in 2010 as an improvement of Das et al.'s scheme by introducing hashed password and mutual authentication between the sensor node and the gateway node by using pre-shared keys, however Vaidya et al. [16] proved that their scheme was vulnerable to impersonation and stolen smart-card attack. In 2012, Ashok et al. [17] proposed a lightweight smart-card based user authentication scheme that provided mutual authentication, key agreement, password protection and allowed a dynamic node addition phase and also was resilient against various attacks. This scheme only uses XOR and hash computation and thus appropriate for WSN however, because of many design flaws and susceptible to various attacks as shown by Shuaiwen and Xiaoming [18] and Turkanovic and Holbl [19], it is infeasible for implementation.

In 2014, Muhamed et al. [20] proposed a new protocol for heterogeneous wireless sensor networks using only XOR and hash functions which was based entirely on smart card authentication technique. This proposed scheme claimed mutual authentication, user anonymity, and energy efficiency. However, this protocol had several security flaws like stolen-verifier, user impersonation, and stolen smart-card attack and also failed to ensure backward secrecy as shown in [21–23]. In 2016, Chang and Le [22] presented the advancement in the scheme which provided perfect forward secrecy but at high storage cost. Farash et al. [21] also proposed an enhanced authentication and key establishment scheme but was again cannot withstand user anonymity problem and attacks such as user impersonation, and stolen smart-card attack as identified by Amin et al. [24]. Later, Amin et al.'s scheme was shown to be vulnerable to Denial-of-Service (DoS) and replay attack by Arasteh et al. [25].

In 2016, Gope and Hwang [26] proposed a realistic anonymous authentication protocol in Wireless sensor networks providing security features like user anonymity, untraceability, perfect forward secrecy, etc. however in 2017, Adavoudi-Jolfaei et al. [27] proved that Gope et al.'s protocol is susceptible to disclosure of session key hence, not providing secure user authentication. In 2017, Li et al. [28] proposed an anonymous mutual authentication scheme for a centralized 2-hop wireless BAN providing anonymous and unlinkable features for sensing devices but Koya and P. P. [29] found key escrow problem in their scheme and also showed their scheme was vulnerable to both sensor and hub node impersonation at-

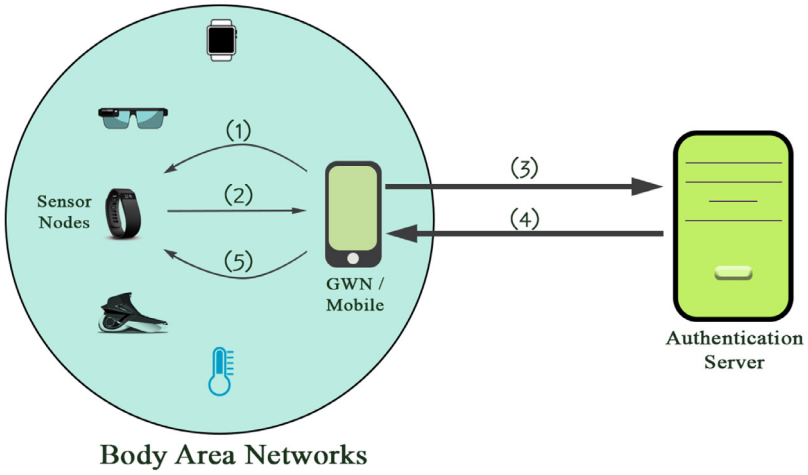


Fig. 1. Five-step authentication model.

tacks. Wu et al. [30] also proposed authentication scheme for wearable devices with the help of cloud server. However, their scheme takes high communication cost in pairing and authenitcation process. Das et al. [31] also designed an authentication protocol for wearable devices using biometrics as an additional feature, but at the cost of high computation overhead, also their algorithm does not consider communication between the cloud server and mobile terminal as a lightweight feature.

Other Researchers have also proposed many asymmetric based cryptographic techniques using RSA, ElGamal, ECC, Paillier cryptosystem, digital signature standards, etc. These public key schemes were highly computational and had memory overheads. Though it is possible to use asymmetric based schemes in IoT Environment, however, performance is the main concern in wireless body area networks.

By the above analysis, we found that though many lightweight authentication protocols have been proposed in the past; However none of them satisfies both lightweight functionality and provide high security with user anonymity, perfect forward secrecy, resilience against Replay, privilege-insider and man-in-the-middle attack. This motivates us to design a protocol which is lightweight and provide secure communication of private data with all the above functionalities.

4. System model

This Section introduces the two models followed in the proposed scheme.

4.1. Network model

An exemplary network model for a wireless body area networks (WBAN) is presented in the Fig. 1. It consists of three types of entities, i.e., an Authentication server, the gateway/mobile terminal, and the wearable sensing devices. The wearable devices, directly attached to the user, sense the health data and forward to the gateway/mobile terminal where a user can access it. The mobile terminal shares the user's health data with the cloud server/Trusted Authority. The Authentication server stores the sensitive data and also helps in mutual authentication between a wearable sensing device and the gateway/mobile terminal. The wearable devices are connected directly to the mobile terminal but are out of range from the cloud server. Therefore, they communicate with each other the help of the gateway/mobile terminal.

4.2. Threat model

We adopt the well known Dolev–Yao threat model [32] where each entity involve in a communication transmits the message over an insecure channel. Under the Dolev–Yao threat model-

- An adversary $\mathbb{A}$ knows an authentication protocol used and may eavesdrop all the messages transmitted over an insecure channel.
- An adversary $\mathbb{A}$ may also modify, corrupt, delete, or redirect the transmitted messages or replay the eavesdropped messages.
- In addition, an adversary $\mathbb{A}$ may also physically capture any sensing device and can extract all the stored parameters from its memory using the power analysis attack [33].
- However, since the mobile terminal is with the user all time, it is assumed that the mobile terminal cannot be captured by an adversary $\mathbb{A}$.

5. Proposed scheme

In this section, we present a new lightweight anonymous user authentication and key-establishment protocol for wearable devices. This scheme is based on five-step authentication model where a user initiates an authentication phase by contacting a required wearable sensing device. Both wearable sensing device and the mobile terminal establishes a session key by means of trusted server for their secure communication. The proposed protocol uses only secure cryptographic hash functions and bitwise XOR operations, hence it is lightweight. It also uses random nonces and current timestamps to resist the strong replay attack. Table 1 displays the list of notations used throughout the proposed scheme.

Our protocol has the following four phases namely, (1) system setup, (2) registration, (3) authentication and key-establishment, and (4) password-change. Here is the brief description of the phases used in proposed scheme-

- System setup phase: In this phase, both the wearable sensing device SD_j and the gateway/mobile terminal GWN_i are preloaded with their secret parameters in their memory as discussed in the Section 5.1.
- Registration phase: This phase is divided into two parts, first the user U_i registers himself/herself to the cloud server. The server stores the credentials of the user in the gateway/mobile terminal's memory needed for the authentication between SD_j and GWN_i. After the user registration, wearable sensing device also registers to the cloud server and associates itself to the mobile terminal as described in the Section 5.2.

Table 1
Notations used throughout the paper.

Symbol	Description
U_i	ith User
SD_j	jth Sensing Device
GWN_i	ith Gateway/Mobile Terminal
TA	Trusted Authority/Cloud Server
ID_u	Identity of a User
SID_j	Identity of jth Sensing Device
GID_i	Identity of ith Mobile Terminal
$MSID_j$	Masked Identity of jth Sensing Device
$MGID_i$	Masked Identity of ith Mobile Terminal
PW_u	Password of the User
X_{GD_i}	Shared key b/w GWN_i and Cloud Server
X_{SD_j}	Shared key SD_j and Cloud Server
X_{Ser}	Secret key of Cloud Server
T_i	Current Timestamp
ΔT	Maximum transmission delay
SK_{ij}	Session-key b/w GID_i and SID_j
$\oplus$	Bitwise XOR Operation
h(·)	Cryptographic hash function
$\|\|$	Concatenation Operation

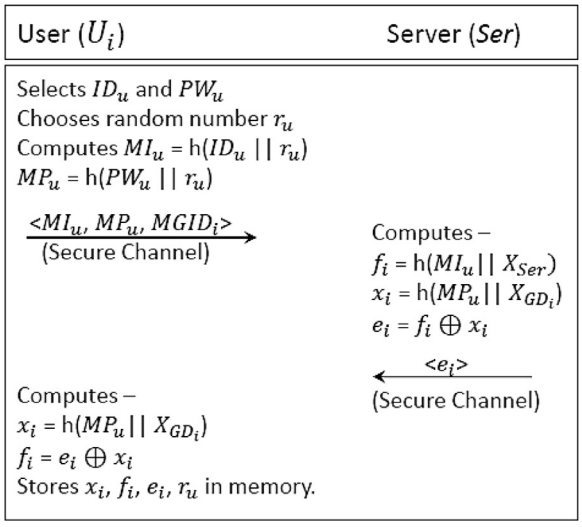


Fig. 2. User registration phase.

- Authentication and key-establishment phase: After the devices are successfully registered, this step is executed for achieving the mutual authentication between SD_j and GWN_i. GWN_i requests SD_j for authentication verification. With the help of the cloud server, both SD_j and GWN_i authenticates each other and generates a temporary session key for further communication as discussed in the Section 5.3.
- Password-change phase: It is an optional phase required when the user needs to change his/her old password. First the user must login with the old credentials to verify his/her identity, then he/she is required to enter a new password and all the credentials are updated accordingly as discussed in the Section 5.4.

We give the detailed description of all the phases next in the subsequent sections.

5.1. System setup phase

The system setup phase is divided into two parts-

5.1.1. Sensing device setup phase

- Trusted Authority (*TA*) registers each sensing device before their deployment.
- *TA* chooses a unique identity SID_j and a long-term shared-key X_{SD_j} for each device.
- TA then computes Masked Identity $MSID_j$ of a sensing device as $MSID_j = h(SID_j \| X_{SD_j})$ and pre-loads $(h(\cdot), SID_j, X_{SD_j}, MSID_j)$ in the memory of SD_j.
- TA stores SID_j, $MSID_j$ and X_{SD_j} in its database as a reference for the jth wearable sensing device.

5.1.2. Mobile terminal setup phase

- For each gateway/mobile terminal GWN_i, Trusted Authority (*TA*) chooses a unique identity GID_i and a long-term shared-key X_{GD_i}.
- *TA* then computes Masked Identity $MGID_i$ of gateway/mobile terminal as $MGID_i = h(GID_i \| X_{GD_i})$ and stores in the memory of gateway node.
- The gateway now has $(h(\cdot), GID_i, X_{GD_i}, MGID_i)$ in its memory.
- TA stores GID_i, $MGID_i$ and X_{GD_i} in its database as a reference for the ith gateway/mobile terminal.

5.2. Registration phase

Two separate phases are required for the registration of the user and the wearable sensing device. The first is to register the user with the cloud server, and second is to register the wearable sensing device to the cloud server using mobile terminal as a gateway node.

5.2.1. User registration phase

The user U_i who owns the mobile terminal executes the following steps-

- U_i chooses a unique identity ID_u and a password PW_u.
- GWN_i chooses a random number r_u to compute a masked identity MI_u of U_i as $MI_u = h(ID_u \| r_u)$.
- *GWN* also computes a masked password MP_u as $MP_u = h(PW_u \| r_u)$ using the same random number r_u.
- *GWN* sends the registration request message containing the masked identity MI_u and the masked password MP_u of the user U_i, and the masked identity $MGID_i$ of the gateway/mobile terminal GWN_i to the cloud server via a secure channel.
- Upon receiving this message from the user U_i, the cloud server computes the secret credentials for the user U_i as $f_i = h(MI_u \| X_{ser})$ using the received masked identity MI_u and its own secret key X_{ser}, and $x_i = h(MP_u \| X_{GD_i})$ using the received masked password MP_u and the shared secret key with GWN_i.
- After calculating f_i and x_i, the cloud server calculates $e_i = f_i \oplus x_i$ and sends e_i to the gateway/mobile terminal GWN_i via a secure channel.
- Upon receiving e_i from the cloud server, GWN_i computes $x_i = h(MP_u \| X_{GD_i})$ using its own saved credentials and calculates $f_i = e_i \oplus x_i$ and stores x_i, f_i, e_i, and r_u in its memory.

Fig. 2 shows the user registration process with the cloud server.

5.2.2. Sensing device registration phase

This phase registers the wearable sensing device to the cloud server with the help of the gateway/mobile terminal. Fig. 3 shows the wearable sensing device registration process with the cloud server. In this phase, the following steps are performed-

- The wearable sensing device SD_j chooses a random number rm_j to compute MP_j as $MP_j = h(SID_j \| X_{SD_j} \| rm_j \| T_1)$ using the current timestamp T_1. SD_j then computes a masked nonce MN_j as $MN_j = X_{SD_j} \oplus rm_j$ and sends $MSID_j$, MN_j, MP_j, and T_1 to the gateway/mobile terminal GWN_i via an unsecure channel.
- Upon receiving this message from SD_j, GWN_i first checks whether $|T_1 - T_2| < \Delta T$ or not, where T_2 is the current timestamp of GWN_i and T_1 is the received timestamp from SD_j. If

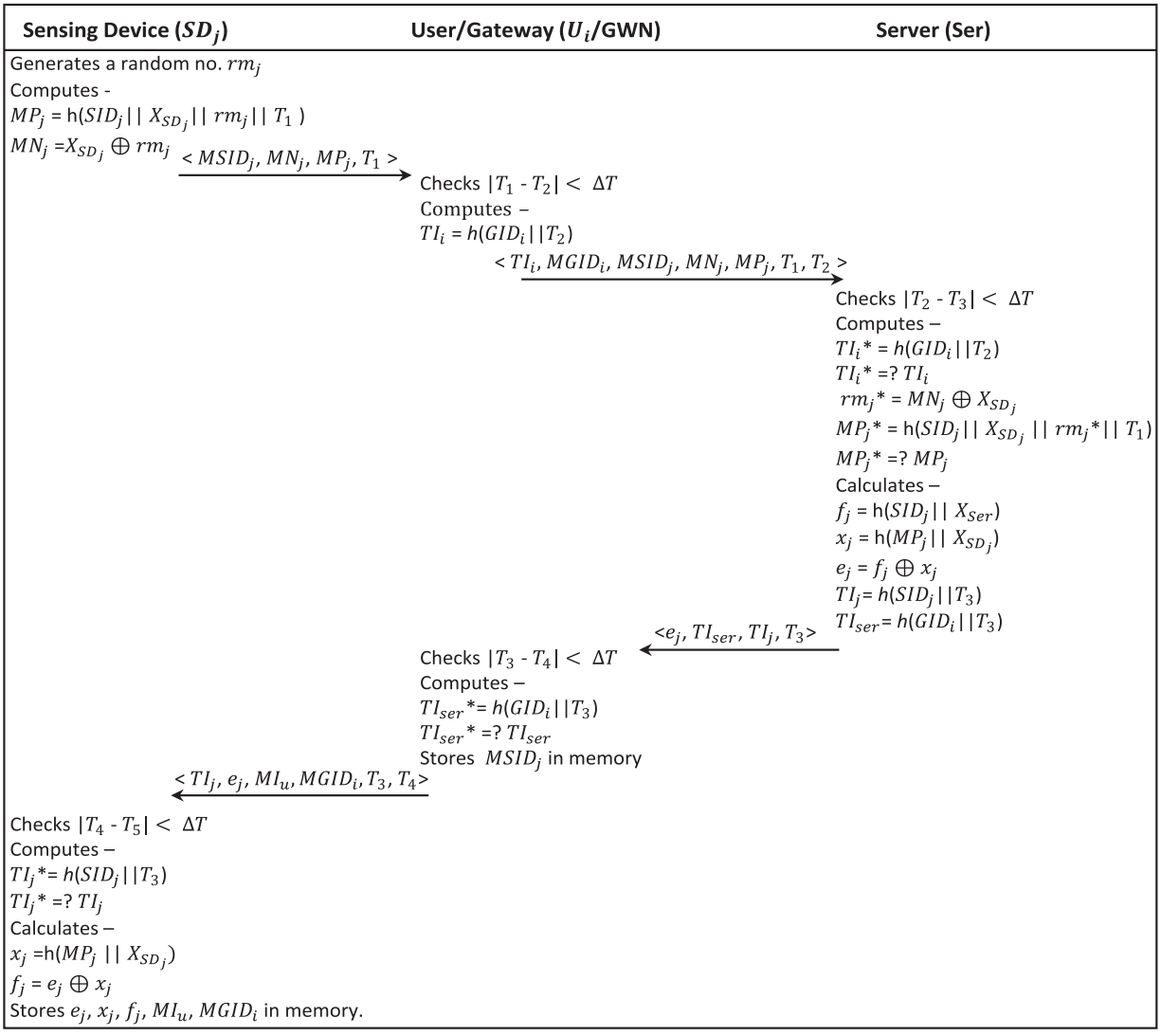


Fig. 3. Sensing device registration phase.

the time difference is less than the maximum allowed transmission delay, GWN_i computes $TI_i = h(GID_i \| T_2)$ and forwards TI_i, $MGID_i$, $MSID_j$, MN_j, MP_j, T_1 and T_2 to the cloud server via an unsecure channel, otherwise it aborts the operation.

- Upon receiving this message, the cloud server checks whether $|T_2 - T_3| < \Delta T$ or not, where T_3 is the current timestamp of the cloud server. If the time delay is more than the expected delay, the cloud server aborts the further operation. Otherwise, it computes $TI_i^* = h(GID_i \| T_2)$ and checks whether $TI_i^* = TI_i$ or not to confirm the message is received from GWN_i. If both are same, it computes $rm_j^* = MN_j \oplus X_{SD_j}$ and $MP_j^* = h(SID_j \| X_{SD_j} \| rm_j^* \| T_1)$ and checks whether $MP_j^* = MP_j$ or not to confirm the message is from SD_j. Again if both are same, it computes the secret credentials for SD_j as $f_j = h(SID_j \| X_{ser})$ using the stored identity SID_j and its own secret key X_{ser}, $x_j = h(MP_j \| X_{SD_j})$ using the received MP_j and the shared secret key with SD_j, and $e_j = f_j \oplus x_j$. It also computes $TI_j = h(SID_j \| T_3)$ and $TI_{ser} = h(GID_i \| T_3)$ and sends e_j, TI_{ser}, TI_j and T_3 to GWN_i via an open channel.
- GWN_i checks whether $|T_3 - T_4| < \Delta T$ or not, where T_4 is the current timestamp of GWN_i. If the time difference is less than the maximum allowed transmission delay, it computes $TI_{ser}^* = h(GID_i \| T_3)$ and checks whether $TI_{ser}^* = TI_{ser}$ or not to confirm the message is coming from the cloud server. If both are same, it stores $MSID_j$ in its own memory and sends TI_j, e_j, MI_u, $MGID_i$, T_3 and T_4 to SD_j, otherwise it aborts the operation.
- Upon receiving this message, SD_j checks whether $|T_4 - T_5| < \Delta T$ or not, where T_5 is the current timestamp of SD_j. If it holds, SD_j computes $TI_j^* = h(SID_j \| T_3)$ and checks whether $TI_j^* = TI_j$ or not to confirm the message is coming from the cloud server. If both are same, it computes $x_j = h(MP_j \| X_{SD_j})$, $f_j = e_j \oplus x_j$ and stores e_j, x_j, f_j, MI_u, and $MGID_i$ in its memory.

5.3. Authentication and key-establishment phase

This phase executes the subsequent steps to attain the mutual authentication between the wearable sensing device and the gateway/mobile terminal. Fig. 4 shows an illustration of the authentication and key-establishment phase.

- The user U_i first inputs his/her identity ID_u^* and password PW_u^* into the gateway/mobile terminal GWN_i for login purpose.
- GWN_i computes $MI^*{}_u = h(ID_u \| r_u)$, $MP_u^* = h(PW_u^* \| r_u)$, $x_i^* = h(MP_u^* \| X_{GD_i})$ and checks whether MI_u^* and x_i^* are equal to MI_u and x_i respectively or not. If they are equal, then the user has entered the correct identity and password, GWN_i then requests SD_j for authentication by sending the user's masked identity

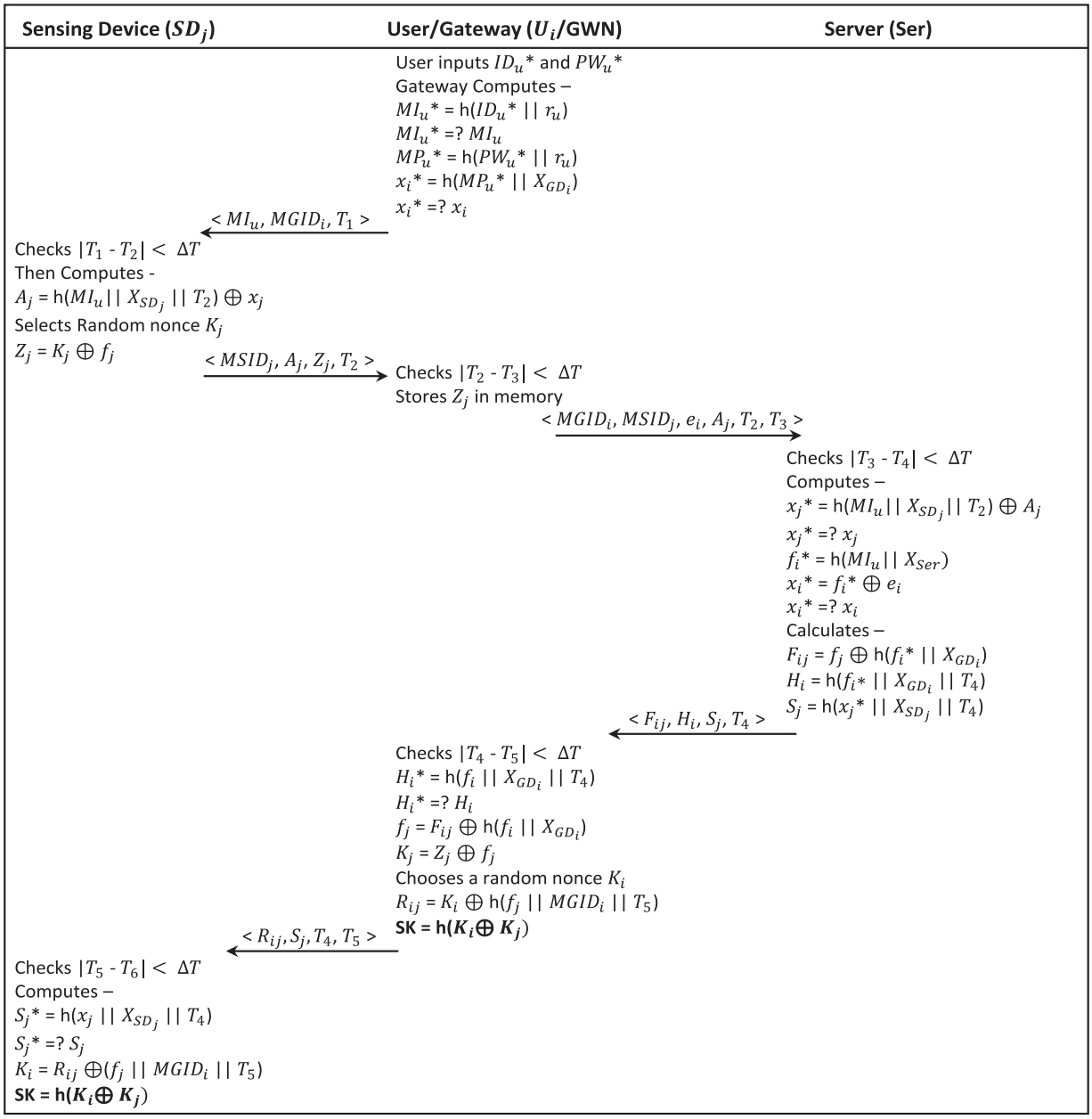


Fig. 4. Authentication and key-establishment phase.

MI_u, its own masked identity $MGID_i$ and the current timestamp T_1.

- The wearable sensing device SD_j first checks the validity of the timestamp, i.e., $|T_1 - T_2| < \Delta T$ or not, where T_2 is SD_j's current timestamp. If the value is less than the maximum transmission delay time, SD_j generates a random nonce k_j and computes $A_j = h(MI_u \parallel X_{SD_j} \parallel T_2) \oplus x_j$ and $Z_j = k_j \oplus f_j$ and sends its $MSID_j$, A_j, Z_j, and T_2 to GWN_i, otherwise it aborts the connection.
- Upon receiving this message from SD_j, GWN_i verifies the timestamp validity, i.e., $|T_2 - T_3| < \Delta T$ or not, where T_3 is the current timestamp of GWN_i. If it is valid, GWN_i stores Z_j in its memory and sends $MGID_i$, $MSID_j$, e_i, A_j, T_2 and T_3 to the cloud server, otherwise it aborts the connection.
- Upon receiving the message, the cloud server checks the timestamp validity, i.e. $|T_3 - T_4| < \Delta T$ or not, where T_4 is the timestamp generated by the server. If it is not valid, the server aborts the further operation; otherwise it calculates x_j^* as $x_j^* = A_j \oplus h(MI_u \parallel X_{SD_j} \parallel T_2)$ and compares it with the x_j stored in its memory for the verification of the wearable sensing device SD_j.
- After successfully verifying SD_j, the cloud server computes f_i^* as $f_i^* = h(MI_u \parallel X_{ser})$, $x_i^* = e_i \oplus f_i^*$ and compares x_i^* with x_i for user verification.
- After successfully verifying the user, the cloud server computes $F_{ij} = f_j \oplus h(f_i^* \parallel X_{GD_i})$, $H_i = h(f_i^* \parallel X_{GD_i} \parallel T_4)$, and $S_j = h(x_j^* \parallel X_{SD_j} \parallel T_4)$ and sends them to the GWN_i with the current timestamp T_4.
- Upon receiving the message from the cloud server, GWN_i first verifies the timestamp validity, i.e., $|T_4 - T_5| < \Delta T$ or not, where T_5 is GWN_i's current timestamp. If it holds, then GWN_i calculates $H_i^* = h(f_i \parallel X_{GD_i} \parallel T_4)$ and compares it with the received H_i from the server. If both are same, then it computes $f_j = F_{ij} \oplus h(f_i \parallel X_{GD_i})$ and $K_j = Z_j \oplus f_j$; otherwise, aborts the connection.

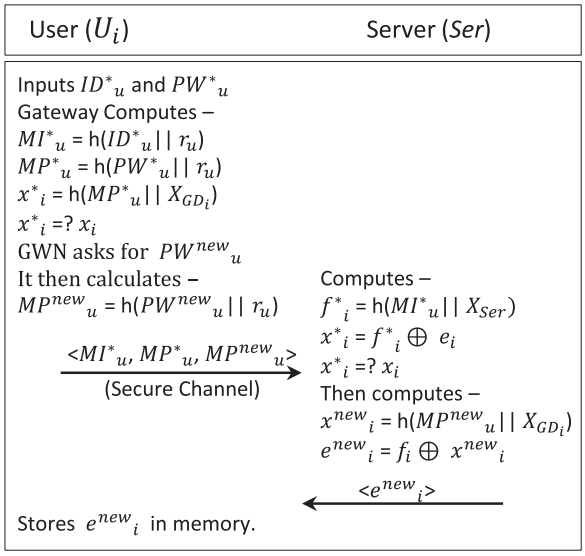


Fig. 5. Password-change phase.

- The gateway/mobile terminal GWN_i then generates a random nonce K_i and computes $R_{ij} = h(f_j \| MGID_i \| T_5) \oplus K_i$ and $SK = h(K_i \oplus K_j)$, where SK is the temporary session key. It then sends R_{ij}, S_j, T_4, and T_5 to SD_j.
- The wearable sensing device SD_j first checks the timestamp validity, i.e., $|T_5 - T_c| < \Delta T$ or not. If it holds, then SD_j calculates $S^*_j = h(x_j \| X_{SD_j} \| T_4)$ and compares it with the received S_j.
- If S^*_j is equal to the S_j, then SD_j verifies the message that it has been sent by the server. Therefore, it computes $K_i = R_{ij} \oplus h(f_j \| MGID_i \| T_5)$ and from K_i, it derives the session key as $SK = h(K_i \oplus K_j)$.

5.4. Password-change phase

This optional phase is needed when the user U_i wants to update his/her old password. Fig. 5 depicts an illustration of the password-change phase. The following steps are to be performed in this phase.

- The user U_i inputs its identity ID^*_u and password PW^*_u into the mobile terminal.
- GWN_i computes MP^*_u using inputted password PW^*_u and the random number r_u.
- GWN_i then computes x^*_i and compares it with original x_i. If they are not equal, the inputted password is not correct and GWN_i aborts the further operation.
- If they are equal, GWN_i asks the user to input new password PW^{new}_u and then calculates the new masked password $MP^{new}_u = h(PW^{new}_u \| r_u)$ and sends MI^*_u, MP^*_u and MP^{new}_u to the cloud server.
- Server computes x^*_i using MP^*_u and stored X_{GDi} and compares it with x_i. If they are equal, then Server computes $x^{new}_i = h(MP^{new}_u \| X_{GDi})$ and $e^{new}_i = f_i \oplus x^{new}_i$ and sends e^{new}_i to the GWN_i.
- GWN_i updates e^{new}_i in its memory.

6. Security analysis of the proposed protocol

In this section, we provide the detailed security analysis of our proposed protocol using the formal security verification techniques and the informal security analysis. First, we use the BAN-logic [34] to prove that our protocol establishes a secure session key between the mobile terminal and the wearable sending device. Second, we use the AVISPA security analyzer tool [35] to prove that our protocol is secure and resistible to different known attacks such as man-in-the-middle and replay attack. Finally, we discuss the security against different popular attacks informally.

6.1. Mutual authentication using BAN-Logic

This section gives the detailed description of the security analysis of our scheme using Burrows–Abadi–Needham (BAN)Logic [34]. The following basic logical notations are used to analyze this scheme:

- $A| \equiv S$: A believes the statement S
- $A \triangleleft S$: A sees the statement S
- $A| \sim S$: A once said S
- $A| \Rightarrow S$: A has jurisdiction over S
- $\#(S)$: S is a fresh statement
- $A \overset{K}{\leftrightarrow} B$: K is a shared secret key between A and B
- $\{S\}_K$: S is encrypted with key K
- $(S)_K$: S is hashed with key K
- (S, Y): S or Y is one part of the formula (S, Y)

The following rules are used in the BAN-logic to describe the main logical postulates:

6.1.1. Message meaning rule

If A sees a statement S encrypted with key K and A believes K is a shared secret key between A and B, then A believes B once said S.

$$\frac{A \mid\equiv A \overset{k}{\leftrightarrow} B, A \triangleleft \{S\}_k}{A \mid\equiv B \mid\sim S}$$

6.1.2. Nonce verification rule

If A believes that the statement S is fresh and A also believes that B once said S, then A believes B believes the statement S.

$$\frac{A \mid\equiv \#\{S\}, A \mid\equiv B \mid\sim S}{A \mid\equiv B \mid\equiv S}$$

6.1.3. Jurisdiction rule

If A believes B has jurisdiction over the statement S and A believes B believes the statement S, then A believes the statement S.

$$\frac{A \mid\equiv B \mid\Rightarrow S, A \mid\equiv B \mid\equiv S}{A \mid\equiv S}$$

6.1.4. Freshness rule

If A believes the part of the statement S is fresh, then A believes that the statement $\{S, Y\}$ is fresh.

$$\frac{A \mid\equiv \#(S)}{A \mid\equiv \#(S, Y)}$$

6.1.5. Belief rule

If A believes B believes the statement (S, Y), then A believes B believes the part of the statement S.

$$\frac{A \mid\equiv B \mid\equiv (S, Y)}{A \mid\equiv B \mid\equiv S}$$

Our main goal is to prove:

- Goal 1: $G_i \mid\equiv (G_i \overset{SK}{\leftrightarrow} S_j)$
- Goal 2: $G_i \mid\equiv S_j \mid\equiv (G_i \overset{SK}{\leftrightarrow} S_j)$
- Goal 3: $S_j \mid\equiv (G_i \overset{SK}{\leftrightarrow} S_j)$
- Goal 4: $S_j \mid\equiv G_i \mid\equiv (G_i \overset{SK}{\leftrightarrow} S_j)$

In our scheme, the messages sent over the un-secure channel are:

- Message 1: $G_i \rightarrow S_j$: $< MI_u, MGID_i, T_1 >$
- Message 2: $S_j \rightarrow G_i$: $< MSID_j, A_j, Z_j, T_2 >$
- Message 3: $G_i \rightarrow Ser$: $< MGID_i, MSID_j, e_i, A_j, Z_j, T_2, T_3 >$
- Message 4: $Ser \rightarrow G_i$: $< F_{ij}, H_i, S_j, T4 >$
- Message 5: $G_i \rightarrow S_j$: $< S_j, R_{ij}, T_4, T_5 >$

To prove the session key establishment between the mobile terminal and the wearable device in the proposed protocol, the following assumptions are made :

- A1: $G_i \mid\equiv \#(T_i), G_i \mid\equiv \#(T_j), G_i \mid\equiv \#(T_{Ser})$
- A2: $S_j \mid\equiv \#(T_i), S_j \mid\equiv \#(T_j)$
- A3: $Ser \mid\equiv \#(T_i), Ser- \mid\equiv \#(T_{Ser})$
- A4: $G_i \mid\equiv (G_i \overset{X_{GDi}}{\longleftrightarrow} Ser)$
- A5: $Ser \mid\equiv (G_i \overset{X_{GDi}}{\longleftrightarrow} Ser)$
- A6: $S_j \mid\equiv (S_j \overset{X_{SDj}}{\longleftrightarrow} Ser)$
- A7: $Ser \mid\equiv (S_j \overset{X_{SDj}}{\longleftrightarrow} Ser)$
- A8: $G_i \mid\equiv S_j \mid\Rightarrow S_j \mid\sim X$, $S_j \mid\equiv G_i \mid\Rightarrow G_i \mid\sim X$

Now, We will analyze the proposed protocol using BAN-logic rules, postulates and assumptions to claim that G_i and S_j share a secret session key for effective communication.

We elaborate our Message 1 as-

$$G_i \rightarrow S_j :< h(r_u \parallel ID_u), h(GID_i \parallel X_{GDi}), T_1 >$$

From Message 1, we get-

$$S_j \triangleleft < (r_u \parallel ID_u), (GID_i \parallel X_{GDi}), T_1 >_{G_i \overset{MSID_j}{\longleftrightarrow} S_j} \quad (1)$$

Using Eq. (1) and MMR, we get-

$$S_j \mid\equiv G_i \mid\sim < (r_u \parallel ID_u), (GID_i \parallel X_{GDi}), T_1 > \quad (2)$$

Using A2, FR and Eq. (2), we get-

$$S_j \mid\equiv \#\{(r_u \parallel ID_u), (GID_i \parallel X_{GDi})\} \quad (3)$$

Using NVR and Eq. (3), we get-

$$S_j \mid\equiv G_i \mid\equiv \{(r_u \parallel ID_u), (GID_i \parallel X_{GDi})\} \quad (4)$$

Next, we elaborate our Message 2 as-

$$S_j \rightarrow G_i :< h(SID_j \parallel X_{SDj}), h(MI_u \parallel X_{SD_j} \parallel T_2) \oplus h(MP_j \parallel X_{SD_j}), k_j \oplus h(SID_j \parallel X_{ser}), T_2 >$$

From Message 2, we get-

$$G_i \triangleleft < (SID_j \parallel X_{SDj}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j}), k_j \oplus h(SID_j \parallel X_{ser}), T_2 > \quad (5)$$

Using MMR and Eq. (5), we get-

$$G_i \mid\equiv S_j \mid\sim < (SID_j \parallel X_{SDj}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j}), k_j \oplus h(SID_j \parallel X_{ser}), T_2 > \quad (6)$$

Using A1,FR,NVR and Eq. (6), we get-

$$G_i \mid\equiv S_j \mid\equiv \{(SID_j \parallel X_{SDj}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j}), k_j \oplus h(SID_j \parallel X_{ser})\} \quad (7)$$

Using M7, BR, JR and Eq. (7), we get-

$$G_i \mid\equiv \{(SID_j \parallel X_{ser}) \oplus k_j\} \quad (8)$$

Now, we elaborate our Message 3 as-

$$G_i \rightarrow Ser :< h(GID_i \parallel X_{GDi}), h(SID_j \parallel X_{SDj}), h(MI_u \parallel X_{ser}) \oplus h(MP_u \parallel X_{GD_i}), h(MI_u \parallel X_{SD_j} \parallel T_2) \oplus h(MP_j \parallel X_{SD_j}), T_2, T_3 >$$

From Message 3, we get-

$$Ser \triangleleft < (GID_i \parallel X_{GDi}), (SID_j \parallel X_{SDj}), (MI_u \parallel X_{ser}) \oplus (MP_u \parallel X_{GD_i}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j}), T_2, T_3 > \quad (9)$$

Using MMR and Eq. (9), we get-

$$Ser \mid\equiv G_i \mid\sim < (GID_i \parallel X_{GDi}), (SID_j \parallel X_{SDj}), (MI_u \parallel X_{ser}) \oplus (MP_u \parallel X_{GD_i}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j}), T_2, T_3 > \quad (10)$$

Using A3, NVR, FR and Eq.(10), we get-

$$Ser \mid\equiv G_i \mid\equiv \{(MI_u \parallel X_{ser}) \oplus (MP_u \parallel X_{GD_i}), (MI_u \parallel X_{SD_j} \parallel T_2) \oplus (MP_j \parallel X_{SD_j})\} \quad (11)$$

Now, we elaborate our Message 4 as-

$$Ser \rightarrow G_i :< h(SID_j \parallel X_{Ser}) \oplus h(h(MI_u \parallel X_{Ser}) \parallel X_{GD_i}), h(h(MI_u \parallel X_{Ser}) \parallel X_{GD_i} \parallel T_4), h(h(MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T4 >$$

From Message 4, we get-

$$G_i \triangleleft < (SID_j \parallel X_{Ser}) \oplus ((MI_u \parallel X_{Ser}) \parallel X_{GD_i}), ((MI_u \parallel X_{Ser}) \parallel X_{GD_i} \parallel T_4), ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T4 > \quad (12)$$

Using MMR and Eq. (12), we get-

$$G_i \mid\equiv Ser \mid\sim < (SID_j \parallel X_{Ser}) \oplus ((MI_u \parallel X_{Ser}) \parallel X_{GD_i}), ((MI_u \parallel X_{Ser}) \parallel X_{GD_i} \parallel T_4), ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T4 > \quad (13)$$

Using A1, NVR, FR and Eq. (13), we get-

$$G_i \mid\equiv Ser \mid\equiv \{(SID_j \parallel X_{Ser}) \oplus ((MI_u \parallel X_{Ser}) \parallel X_{GD_i}), ((MI_u \parallel X_{Ser}) \parallel X_{GD_i} \parallel T_4), ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4)\} \quad (14)$$

Using BR, JR and Eq. (14), we get-

$$G_i \mid\equiv \{(SID_j \parallel X_{Ser}) \oplus ((MI_u \parallel X_{Ser}) \parallel X_{GD_i}), ((MI_u \parallel X_{Ser}) \parallel X_{GD_i} \parallel T_4)\} \quad (15)$$

G_i chooses K_i and generates SK using R_{ij}, Therefore-

$$G_i \mid\equiv (G_i \overset{SK}{\longleftrightarrow} S_j) \longrightarrow \textbf{Goal 1}$$

G_i sends Message 5 to the S_j, Therefore-

$$G_i \mid\equiv S_j \mid\equiv (G_i \overset{SK}{\longleftrightarrow} S_j) \longrightarrow \textbf{Goal 2}$$

We elaborate our Message 5 as-

$$G_i \rightarrow S_j :< h(h(SID_j \parallel X_{Ser}) \parallel MGID_i \parallel T_5) \oplus K_i, h(h(MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T_4, T_5 >$$

From Message 5, we get-

$$S_j \triangleleft < ((SID_j \parallel X_{Ser}) \parallel MGID_i \parallel T_5) \oplus K_i, ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T_4, T_5 > \quad (16)$$

Using MMR and Eq. (16), we get-

$$S_j \mid\equiv G_i \mid\sim < ((SID_j \parallel X_{Ser}) \parallel MGID_i \parallel T_5) \oplus K_i, ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4), T_4, T_5 > \quad (17)$$

Using A2, NVR, FR and Eq. (17), we get-

$$S_j \mid\equiv G_i \mid\equiv \{((SID_j \parallel X_{Ser}) \parallel MGID_i \parallel T_5) \oplus K_i, ((MP_j \parallel X_{SDj}) \parallel X_{SD_j} \parallel T_4)\} \quad (18)$$

Using BR, JR and Eq. (18), we get-
$S_j \mid\equiv \{((SID_j \parallel X_{Ser}) \parallel MGID_i \parallel T_5) \oplus K_i\}$
Therefore,

$$S_j \mid\equiv G_i \mid\equiv (G_i \overset{SK}{\longleftrightarrow} S_j) \longrightarrow \textbf{Goal 4}$$

S_j calculates K_i and generates SK. Therefore,

$$S_j \mid\equiv (G_i \overset{SK}{\longleftrightarrow} S_j) \longrightarrow \textbf{Goal 3}$$

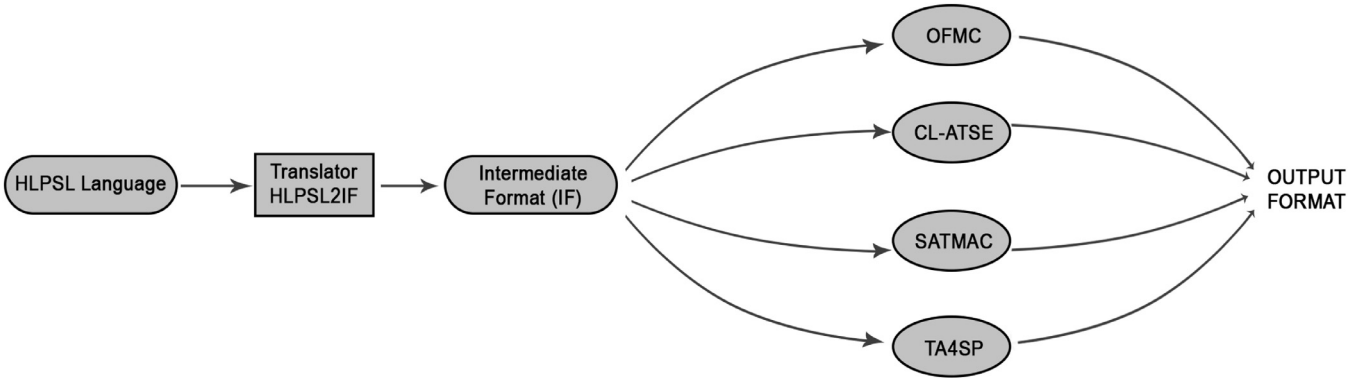


Fig. 6. AVISPA tool architecture.

6.2. Formal security verification using AVISPA tool

This section simulates the proposed protocol for security verification utilizing broadly accepted AVISPA simulation tool [35,36]. AVISPA is a push button formal security verification technique for the automated validation of Internet cryptographic protocols and applications. It measures the security of the protocol by implementing the scheme in High-Level Protocol Specification Language (HLPSL) [37].

HLPSL is a role-oriented model in which each entity plays a role during its execution. There is no dependency among roles. Each role defines the initial parameters and communication channel with the other roles. The attacker uses the Dolev–Yao attack model. The role system also defines the number of sessions, the number of principals and the roles.

The Fig. 6 shows the AVISPA tool architecture. In this architecture, the first step is to preprocess our proposed protocol into HLPSL language which is then given as an input to the HLPSL2IF translator. Using HLPSL2IF translator, the HLPSL language interpretation is translated into an intermediary form (IF). This intermediate format is directly read by the AVISPA tool's back-ends that are integrated with the HLPSL. There are four back-ends implemented in AVISPA simulation tool namely-

- *On-the-fly-Model-Checker (OFMC)* [38] explores the transition system as specified by Intermediate Format in a demand-driven way. It executes several symbolic techniques to detect the attacks by building the infinite tree, and verification of the correctness of the protocol as described by the protocol analysis problem. It supports the typed and un-typed protocol models and algebraic cryptographic operations.
- *Constraint-Logic-based Attack Searcher (CL-AtSe)* [39] translates the security protocol specification in a set of constraints to find the possible attacks on the protocol. CL-AtSe translates and checks the transition relation internally and fully automatically, i.e., no external tools are required to perform these steps. The analysis algorithm is designed to model each protocol step by constraints on the attacker's knowledge for a bounded no.of loops. It applies the redundancy elimination technique and constraint solving to search for the attacks in the full specification for a maximum number of bounded loop iterations.
- *SAT-based Model-checker (SATMC)* [40] unrolls the transition relation specified by IF and encodes it to build a proposition formula which is then given to a state-of-art SAT-solver. It is an open and extensible model checking platform of security protocol specifications to discover attacks on protocols and verify the bounded number of sessions.
- *Tree Automata based on Automatic Approximations for the Analysis of Security Protocols (TA4SP)* uses regular tree automata language to compute an approximation of the intruder's knowledge for the unbounded number of sessions. In an over-approximation context, it may conclude the given initial state is safe for secrecy property and in an under-approximation context, it may show the flaws in the protocol for secrecy property.

6.2.1. Implementation and analysis of results

The implementation for the sensing device registration and the authentication phases are performed using HLPSL specification. It involves three types of roles: (1) *sensor* for the wearable sensing device as depicted in Fig. 7, (2) *user* for the user/gateway as depicted in Fig. 8, and (3) *server* for the trusted cloud server as depicted in Fig. 9. There are total eight messages sent over an insecure channel thereby stating total of nine states in the code. It is to be noted that we have considered the end of the registration phase as the login step for the authentication phase and therefore combining the device registration and the authentication phase in the same code.

The output of all these back-ends have the same format and specify whether the protocol is SAFE or UNSAFE against different attacks such as Replay, impersonation or man-in-the-middle attack and whether ensures secrecy or not. The output format consists of mainly five sections specifying the Summary, Details, Goals, Backend, and Statistics of the protocol.

We have used the SPAN (Security Protocol ANimator for AVISPA) simulator to simulate our proposed protocol. The simulation results under the OFMC back-end and the CL-AtSe back-end are shown in Figs. 10 and 11 respectively. The summary of the results clearly shows that the proposed protocol is safe under different attacks based on Dolev-Yao model. The bitwise XOR operation is not supported by both the SATMC and the TA4SP back-ends at present. Therefore the results shown by these back-ends are inconclusive, and thus we have not included them in this paper.

6.3. Informal security analysis

This section discusses the resilience of the proposed protocol against different well-known security attacks and it also gives the comparison of security features with the other related schemes.

6.3.1. User impersonation attack

Suppose an attacker $\mathbb{A}$ tries to create a valid log-in request. For this, $\mathbb{A}$ has to enter ID_u and PW_u of a user to GWN_j. As the user's identity and password are not shared anywhere through the insecure channel and ID_u cannot be computed from $MI = h(ID_u \| r_u)$ because of one-wayness of hash function, it is computationally infeasible for an attacker $\mathbb{A}$ to guess PW_u and calculate ID_u in a polynomial time. Therefore, the scheme is secure against user impersonation attack.

```
role sensor (SDj, Ui, Ser : agent, SND, RCV : channel(dy), SKsnu, SKusv :symmetric_key, H : hash_func)
played_by SDj
def=
local State : nat,
        Rmj, MPj, MNj, T1, T5, T4, TTIj, Xj, Fj, T2, Aj, Kj, Zj, MSIDj, SIDj : text,
        Rij, TIj, Ej, MIu, MGIDi, T3, XSDj, Sj, T6, TSj, Ki, SK : text
const sp1, sdj_ui_ti : protocol_id
init State:=0
transition
1. State=0 /\ RCV(start) =|> State':=1 /\ Rmj':=new() /\ T1':=new() /\ MPj':=H(SIDj.XSDj.Rmj'.T1 /\ MNj':=xor(XSDj,Rmj') /\
        SND({MSIDj.MNj'.MPj'.T1'}_SKsnu) /\ secret({SIDj,XSDj,MSIDj},sp1,{SDj})
2. State=1 /\ RCV({TIj.Ej.MIu.MGIDi.T3.T4}_SKsnu) =|> State':=2 /\ T5':=new() /\ TTIj':=H(SIDj.T3) /\ request(SDj,Ui,sdj_ui_ti,TTIj') /\
        Xj':=H(MPj.XSDj) /\ Fj':=xor(Ej,Xj') /\ T2':=new() /\ Aj':=xor(H(MIu.XSDj.T2'),Xj) /\ Kj':=new() /\ Zj':=xor(Kj',Fj') /\
        SND({MSIDj.Aj'.Zj'.T2'}_SKsnu)
3. State=2 /\ RCV({Rij.Sj.T4.T5}_SKsnu) =|> State':= 3 /\ T6':=new() /\ TSj':=H(Xj.XSDj.T4) /\ request(SDj,Ui,sdj_ui_sj,TSj') /\
        Ki':=xor(Rij,H(Fj.MGIDi.T5)) /\ SK':=H(xor(Ki',Kj))
end role
```

Fig. 7. HLPSL code for role_sensor.

```
role user(SDj, Ui, Ser : agent, SND, RCV : channel(dy), SKsnu, SKusv : symmetric_key, H : hash_func)
played_by Ui
def=
local State : nat,
        MSIDj, MNj, MPj, T1, T2, T3, TTIser, TIser, MGIDi, GIDi, TIi, MIu, T5, Ej, TIj, T4, Aj : text,
        Zj, Xi, Fij, Hi, Sj, THi, Fi, Xgdi, Fj, Kj, Rij, Sk, Ki : text
const sp1, ser_ui_ti, ui_ser_ti, ser_ui_mp, sp2, sp3, sdj_ui_sj, ser_ui_xi : protocol_id
init State:=0
transition
1. State=0 /\ RCV({MSIDj.MNj.MPj.T1}_SKsnu) =|> State':=1 /\ T2':=new() /\ TIi':=H(GIDi.T2) /\ SND({TIi'.MSIDj.MNj.MPj.T1.T2'}_SKusv) /\
        secret({GIDi},sp1,{Ui}) /\ witness(Ui,Ser,ser_ui_ti,TIi' ) /\ witness(Ui,Ser,ser_ui_mp,MPj)
2. State=1 /\ RCV({Ej.TIser.TIj.T3}_SKusv) =|> State':=2 /\ T4':=new() /\ TTIser':=H(MIu.T3) /\ request(Ui,Ser,ui_ser_ti,TTIser') /\
        SND({TIj.Ej.MIu.MGIDi.T3.T4'}_SKsnu) /\ secret({MGIDi},sp2,{Ui})
3. State=2 /\ RCV({MSIDj.Aj.Zj.T2}_SKsnu) =|> State':= 3 /\ T3':= new() /\ SND({MGIDi.MSIDj.Ej.Aj.T2.T3'}_SKusv) /\
        witness(Ui,Ser,ser_ui_xi,Xi) /\ secret({Xi},sp3,{Ui})
4. State=3 /\ RCV({Fij.Hi.Sj.T4}_SKusv) =|> State':=4 /\ T5':=new() /\ THi':=H(Fi.Xgdi.T4) /\ request(Ui,Ser,ui_ser_hi,THi') /\
        Fj':=xor(Fij,H(Fi.Xgdi)) /\ Kj':=xor(Zj,Fj') /\ Ki':=new() /\ Rij':=xor(Ki',H(Fj'.MGIDi.T5')) /\ Sk':=H(xor(Ki',Kj')) /\
        SND({Rij'.Sj.T4.T5'}_SKsnu) /\ witness(Ui,SDj,sdj_ui_sj,Sj)
end role
```

Fig. 8. HLPSL code for role_user.

```
role server( SDj, Ui, Ser : agent, SND, RCV : channel(dy), SKsnu, SKusv : symmetric_key, H : hash_func)
played_by Ser
def=
local State :nat,
        MGIDi, MSIDj, MNj, MPj, T1, T2, T3, TIser, Xgdi, TTIi, TIi, Rmj, TMPj, Trmj, XSDj, SIDj, Xser, Fj : text,
        Xj, Ej, TIj, MIu, GIDi, Aj, T4, TXj, TFi, TXi, Fij, Hi, Sj : text
const sp1, ser_ui_ti, ser_ui_mp, ui_ser_ti, ui_ser_hi : protocol_id
init State:=0
transition
1. State=0 /\ RCV({MGIDi.TIi.MSIDj.MNj.MPj.T1.T2}_SKusv) =|> State':=1 /\ TTIi':=H(GIDi.T2) /\ request(Ser,Ui,ser_ui_ti,TTIi') /\
        Trmj':=xor(MNj,XSDj) /\ TMPj':=H(SIDj.XSDj.Trmj'.T1) /\ request(Ser,Ui,ser_ui_mp,TMPj') /\ Fj':=H(SIDj.Xser) /\
        Xj':=H(MPj.XSDj) /\ Ej':=xor(Fj',Xj') /\ T3':=new() /\ TIj':=H(SIDj.T3') /\ TIser':=H(MIu.T3') /\ SND({Ej'.TIser'.TIj'.T3'}_SKusv) /\
        secret({XSDj,SIDj,Xser,GIDi},sp1,{Ser}) /\ witness(Ser,Ui,ui_ser_ti,TIser')
2. State=1 /\ RCV({MGIDi.MSIDj.Ej.Aj.T2.T3}_SKusv) =|> State':=2 /\ T4':=new() /\ TXj':=xor(H(MIu.XSDj.T2),Aj) /\ TFi':=H(MIu.Xser) /\
        TXi':=xor(TFi',Ej) /\ request(Ser,Ui,ser_ui_xi,TXi') /\ Fij':=xor(Fj,H(TFi'.Xgdi)) /\ Hi':=H(TFi'.XSDj.T4') /\ Sj':=H(TXj'.XSDj.T4') /\
        SND({Fij'.Hi'.Sj'.T4'}_SKusv) /\ witness(Ser,ui,ui_ser_hi,Hi')
end role
```

Fig. 9. HLPSL code for role_server.

```
% OFMC
% Version of 2006/02/13
SUMMARY
 SAFE
DETAILS
 BOUNDED_NUMBER_OF_SESSIONS
PROTOCOL
 /home/span/span/testsuite/results/myproto.if
GOAL
 as_specified
BACKEND
 OFMC
COMMENTS
STATISTICS
 parseTime: 0.00s
 searchTime: 0.56s
 visitedNodes: 216 nodes
 depth: 6 plies
```

Fig. 10. OFMC back-end simulation result.

```
SUMMARY
 SAFE
DETAILS
 BOUNDED_NUMBER_OF_SESSIONS
 TYPED_MODEL
PROTOCOL
 /home/span/span/testsuite/results/myproto.if
GOAL
 As Specified
BACKEND
 CL-AtSe
STATISTICS
 Analysed  : 9 states
 Reachable  : 9 states
 Translation: 0.04 seconds
 Computation: 0.00 seconds
```

Fig. 11. CL-AtSe back-end simulation result.

6.3.2. Sensing device impersonation attack

Suppose an attacker $\mathbb{A}$ intercepts the transmitted message $< MSID_j, A_j, Z_j, T_2 >$ and tries to generate different message with $< MSID_j, A_j^{adv}, Z_j^{adv}, T_2 >$ by impersonating as a sensing device. $\mathbb{A}$ can generate a random nonce K_j and the current timestamp T_2, however the attacker $\mathbb{A}$ would not be able to create valid message without knowing the secret parameters x_j, X_{SD_j} and f_j and it is computationally infeasible to guess these parameters because of one-wayness of hash function. Therefore, the attacker cannot perform sensing device impersonation attack.

6.3.3. Gateway/mobile terminal impersonation attack

Assume an attacker $\mathbb{A}$ tries to impersonate as a gateway/mobile terminal by intercepting the messages $< MI_u, MGID_i, T_1 >$ and $< MGID_i, MSID_j, e_i, A_j, T_2, T_3 >$. Notice that the parameter f_i is secret and guessing both f_i and shared secret key X_{GD_i} to compute $f_j = F_{ij} \oplus h(f_i \parallel X_G D_i$ is computationally infeasible for $\mathbb{A}$ due to the protection of one-wayness of hash function. Thus, our scheme is secure against this attack.

6.3.4. Eavesdropping attack

In Dolev–Yao threat model, an adversary $\mathbb{A}$ can eavesdrop all the transmitted message over an insecure channel. $\mathbb{A}$ collects the message $< MSID_j, A_j, Z_j, T_2 >$ transmitting from SD_j to GID_i and the message $< R_{ij}, S_j, T_4, T_5 >$ transmitting from GID_i to SD_j. In the proposed scheme, Session Key SK is computed as $SK = h(K_i \oplus K_j)$. $\mathbb{A}$ cannot compute $K_i = R_{ij} \oplus h(f_j \parallel MGID_i \parallel T_5)$ and $K_j = Z_j \oplus f_j$ from the eavesdropped parameters as f_j is protected by cryptographic hash function. Therefore, the scheme is well protected against this attack.

6.3.5. Privileged insider attack

An adversary $\mathbb{A}$ who may be a privileged user tries to send a valid login request and authentication message to SD_j. As the sensing devices are associated to an individual user with their MI_u and mobile terminal calculates the user's MI_u and sends it to SD_j. SD_j calculates $A_j = h(MI_u \parallel X_{SD_j} \parallel T_2) \oplus x_j$ for different user and the cloud server will not validate x_j^* to x_j and aborts the connection. Therefore, this attack is not possible and our scheme is secure against this attack.

6.3.6. Offline password guessing attack

Suppose an attacker $\mathbb{A}$ guesses the user's identity and password in polynomial time to login as a valid user by calculating $MI_u = h(ID_u^{guess} \parallel r_u)$ and $MP_u = h(PW_u^{guess} \parallel r_u)$. The attacker has to capture mobile terminal to perform this attack. As GWN_i is always with the user and because of our assumption in threat model that Mobile Terminal capture attack is not possible, our scheme is protected against offline password guessing attack.

6.3.7. Anonymity and untraceability

We have used the masked identites for user and sensing devices to hide their real identities U_i and SID_j. An adversary $\mathbb{A}$ cannot interpret ID_u and SID_j from MI_u and $MSID_j$ respectively as they are protected by hash property hence preserving user and sensing device anonymity. Furthermore, we have used fresh nonces in each session and current timestamps everywhere, therefore all sessions are unlinkable and an adversary cannot trace any device from previous messages hence preserving untraceability.

6.3.8. Perfect forward and backward secrecy

Assume an adversary $\mathbb{A}$ compromises the session key SK, it must not reveal any previous or future sessions and must not compromise further SK_{ij}. As the SK is calculated as $SK = h(K_i \oplus K_j)$, it does not reveal any other secret parameters and also all sessions use random nonces and current timestamps to compute it. Therefore, our scheme preserves these properties.

6.3.9. Sensing device capture attack

Suppose an attacker $\mathbb{A}$ captures one of the sensing devices and extract all the secret parameters stored in it using power analysis attack. $\mathbb{A}$ will only get the masked identity of a user and masked identity of a gateway/mobile terminal which are already sent over an insecure channel. No other parameters of a user is shared with any SD_j and all sensing devices have different secret shared keys with the cloud server, an attacker cannot compromise any other SK between GID_i and non-compromised SD_j. Hence, the scheme is secure against this attack and performs smoothly.

The comparison of security features of our protocol with the other related schemes is shown in Table 2. In this table, the value *Y* means a protocol satisfies the property or resilient against the attack and the value *N* denotes that a protocol does not satisfy the property or not resilient against the attack. Our protocol provides more security features as compared to the other related existing schemes.

Table 2
Comparison of security features.

Security property	[22]	[24]	[26]	[28]	[30]	[31]	Proposed
Anonymity and untraceability	N	N	N	Y	Y	Y	Y
Perfect forward secrecy	N	Y	Y	Y	N	N	Y
Replay attack	Y	N	Y	Y	Y	Y	Y
User impersonation attack	Y	N	Y	Y	Y	Y	Y
Sensing device impersonation attack	Y	Y	Y	N	Y	Y	Y
Gateway impersonation attack	Y	Y	Y	N	Y	Y	Y
Node capture attack	N	Y	Y	Y	N	N	Y
Offline guessing attack	N	Y	Y	Y	Y	N	Y
Privileged insider attack	N	Y	N	Y	N	Y	Y
Man-in-the-middle attack	Y	Y	Y	Y	Y	Y	Y

Table 3
Computation cost comparison.

Scheme	Sensor side	User side	Server side	Total Cost
Amin et al. [24]	$5T_h + 3T_{XOR}$	$12T_h + 7T_{XOR}$	$15T_h + 7T_{XOR}$	$32T_h + 17T_{XOR}$
Chang and Le [22]	$5T_h + 4T_{XOR}$	$7T_h + 4T_{XOR}$	$8T_h + 1T_{XOR}$	$20T_h + 9T_{XOR}$
Gope and Hwang [26]	$3T_h + 1T_{XOR}$	$14T_h + 7T_{XOR}$	$9T_h + 4T_{XOR}$	$26T_h + 12T_{XOR}$
Adavoudi–Jolfaei et al. [27]	$3T_h + 2T_{XOR}$	$8T_h + 9T_{XOR}$	$9T_h + 7T_{XOR}$	$20T_h + 19T_{XOR}$
Li et al. [28]	$3T_h + 7T_{XOR}$	-	$4T_h + 12T_{XOR}$	$7T_h + 19T_{XOR}$
Wu et al. [30]	$6T_h + 1T_{XOR}$	$7T_h + 1T_{XOR}$	$10T_h + 2T_{XOR}$	$23T_h + 4T_{XOR}$
Das et al. [31]	$7T_h + 2T_{XOR}$	$9T_h + 5T_{XOR}$	-	$16T_h + 7T_{XOR}$
Proposed	$4T_h + 4T_{XOR}$	$7T_h + 4T_{XOR}$	$5T_h + 3T_{XOR}$	$16T_h + 11T_{XOR}$

7. Performance evaluation of the proposed protocol

In this section, we provide a comparative performance analysis of our proposed protocol. We compare our proposed protocol with the other related authentication schemes regarding communication and computational cost. Since in our scheme, a user initiates an authentication process to share the secure session-key, the mobile terminal has to perform more computations than wearable sensing device and the cloud server, however with the comparative evaluation, we show that our scheme still gives better performance as compared to other related schemes regarding computational and communication cost.

7.1. Computation cost

Table 3provides the summary of the Computation cost analysis. We compare the execution time for all protocols on selective similar factors as the component and the composition of other schemes are different from ours. In Table 3, T_h and T_{XOR} stand for the time needed for the hash operations and the XOR operations respectively. In Li et al.'s scheme [28], intermediate node does not perform any computation rather just forward the message received by the sensing device to the Hub node. Therefore, the computation cost at the user/gateway side is left blank in the Table. Also Das et al.'s scheme [31] doesn't focus on the communication between the mobile device and the cloud server, hence the Server side computation is left blank in the Table 3. It is clear from the analysis that our proposed protocol is efficient in terms of computation cost for resource constraint wearable devices.

7.2. Communication cost

To compare the communication overhead among our protocol and the other proposed schemes, we are taking the following assumptions-

- Identity *ID*, passwords, and random nonces are of 128 bits length.
- SHA3-256 Hash function is generating hash digest of length 256 bits.
- Time stamp is of length 32 bits.
- Shared keys are of length 160 bits.

Table 4
Communication cost comparison.

Scheme	No. of msgs.	No. of bits
Amin et al. [24]	6	4096
Chang and Le [22]	4	3104
Gope and Hwang [26]	4	3184
Adavoudi–Jolfaei et al. [27]	4	3696
Li et al. [28]	4	4672
Wu et al. [30]	5	4800 (pairing) 3932 (authentication)
Das et al. [31]	3	1696
Proposed	5	3808

It is to be noted that SHA3-256 hash function is more secure than SHA-1, therefore, the hash digest in our scheme is of 256 bits. For the comparison of the communication overhead with the other related schemes, we have assumed the same hash function and calculated the communication overheads in other schemes.

Table 4 shows the communication cost for the schemes during the login and authentication phases. Our scheme requires less communication overhead as compared to [24,28,30]. Das et al.'s [31] scheme does not show the communication between the mobile device and the cloud server. Though our scheme requires more communication cost as compared to [22,26,27], it is justifiable as our scheme provides more secure features and resistible to attacks which were there in those schemes.

8. Conclusion

In this paper, we have presented a new lightweight anonymous user authentication and key-establishment scheme for wearable devices that ensures the secure communication within the IoT environment. Wearable devices typically consist of low power, low cost, lightweight devices which sends the personal and sensitive information of a user to a mobile device, so we have used only hash functions and bitwise exclusive-OR operations for energy efficiency. The proposed protocol uses the masked identity of the de-

vices to ensure anonymity and privacy preserving. We have proved the mutual authentication between the gateway/mobile terminal and the wearable sensing device using the BAN-logic. We have also shown the security of the protocol using broadly accepted AVISPA simulation tool. Our protocol is cost efficient in terms of computation and communication overheads as compared to the other related schemes. Hence the proposed protocol is suitable for IoT environment and practical applications.

References

[1] S.M.R. Islam, D. Kwak, M.H. Kabir, M. Hossain, K. Kwak, The internet of things for health care: a comprehensive survey, IEEE Access 3 (2015) 678–708, doi:10.1109/ACCESS.2015.2437951.

[2] P. Olli, Introduction to Wearable Healthcare Technology, University of Jyväskylä, 2017, pp. 1–26.

[3] K. Tehrani, A. Michael, Wearable Technology and Wearable Devices: Everything You Need to Know, Wearable Devices Magazine (2014) WearableDevices.com, March. Web..

[4] M. Haghi, et al., Wearable Devices in Medical Internet of Things: Scientific Research and Commercially Available Devices, Healthcare informatics research 23 (1) (2017) 4–15.

[5] P.J. Soh, G.A.E. Vandenbosch, M. Mercuri, D.M.M. Schreurs, Wearable wireless health monitoring: current developments, challenges, and future trends, IEEE Microw. Mag. 16 (4) (2015) 55–70, doi:10.1109/MMM.2015.2394021.

[6] M.A. Ferrag, L.A. Maglaras, H. Janicke, J. Jiang, L. Shu, Authentication Protocols for Internet of Things: A Comprehensive Survey, Security and Communication Networks 2017 (2017) 41 Article ID 6562953, doi:10.1155/2017/6562953.

[7] R. Rivest, The Md5 message-digest algorithm, https://www.rfc-editor.org/rfc/rfc1321.txt1992.

[8] P.J.D. Eastlake, Us secure hash algorithm 1 (sha1), https://www.rfc-editor.org/rfc/rfc3174.txt2001.

[9] H. Dobbertin, A. Bosselaers, B. Preneel, Ripemd-160: a strengthened version of Ripemd, in: Fast Software Encryption, Springer Berlin Heidelberg, Berlin, Heidelberg, 1996, pp. 71–82.

[10] M.J.O. Saarinen, J.P. Aumasson, The Blake2 cryptographic hash and message authentication code (MAC), https://www.rfc-editor.org/rfc/rfc7693.txt2015.

[11] S. jen Chang, R. Perlner, W.E. Burr, M.S. Turan, J.M. Kelsey, S. Paul, L.E. Bassham, Third-round report of the Sha-3 cryptographic hash algorithm competition, NIST Inter. Rep. 7896 (2012).

[12] K.H.M. Wong, Y. Zheng, J. Cao, S. Wang, A dynamic user authentication scheme for wireless sensor networks, in: Proceedings of the IEEE International Conference on Sensor Networks, Ubiquitous, and Trustworthy Computing (SUTC'06), 1, 2006, p. 8pp., doi:10.1109/SUTC.2006.1636182.

[13] H. Tseng, R. Jan, W. Yang, An improved dynamic user authentication scheme for wireless sensor networks, in: Proceedings of the IEEE GLOBECOM IEEE Global Telecommunications Conference, 2007, pp. 986–990, doi:10.1109/GLOCOM.2007.190.

[14] M.L. Das, Two-factor user authentication in wireless sensor networks, IEEE Trans. Wireless Commun. 8 (3) (2009) 1086–1090, doi:10.1109/TWC.2008.080128.

[15] M.K. Khan, K. Alghathbar, Cryptanalysis and security improvements of 'two-factor user authentication in wireless sensor networks', Sensors 10 (3) (2010) 2450–2459. https://10.3390/s100302450; http://www.mdpi.com/1424-8220/10/3/2450

[16] B. Vaidya, D. Makrakis, H.T. Mouftah, Improved two-factor user authentication in wireless sensor networks, in: Proceedings of the IEEE 6th International Conference on Wireless and Mobile Computing, Networking and Communications, 2010, pp. 600–606. https://10.1109/WIMOB.2010.5645004

[17] K.D. Ashok, S. Pranay, C. Santanu, K.S. Jamuna, A dynamic password-based user authentication scheme for hierarchical wireless sensor networks, J. Netw. Comput. Appl. 35 (5) (2012) 1646–1656. https://doi.org/10.1016/j.jnca.2012.03.011; http://www.sciencedirect.com/science/article/pii/S1084804512000926

[18] X. Shuaiwen, W. Xiaoming, A new user authentication scheme for hierarchical wireless sensor networks, Int. Rev. Comput. Soft. (IRECOS) 8 (1) (2013). https://www.praiseworthyprize.org/jsm/index.php? journal=irecospage=articleop=viewpath5B5D=11211

[19] M. Turkanovic, M. Holbl, An improved dynamic password-based user authentication scheme for hierarchical wireless sensor networks, Elektronika ir Elektrotechnika 19 (6) (2013). http://eejournal.ktu.lt/index.php/elt/article/view/2038

[20] T. Muhamed, B. Boštjan, H. Marko, A novel user authentication and key agreement scheme for heterogeneous ad hoc wireless sensor networks, based on the internet of things notion, Ad. Hoc. Netw. 20 (2014) 96–112. https://doi.org/10.1016/j.adhoc.2014.03.009;http://www.sciencedirect.com/science/article/pii/S157087051400064X

[21] S.F. Mohammad, T. Muhamed, K. Saru, H. Marko, An efficient user authentication and key agreement scheme for heterogeneous wireless sensor network tailored for the internet of things environment, Ad. Hoc. Netw. 36 (2016) 152–176. https://doi.org/10.1016/j.adhoc.2015.05.014;http://www.sciencedirect.com/science/article/pii/S1570870515001195

[22] C. Chang, H. Le, A provably secure, efficient, and flexible authentication scheme for ad hoc wireless sensor networks, IEEE Trans. Wireless Commun. 15 (1) (2016) 357–366, doi:10.1109/TWC.2015.2473165.

[23] R. Amin, G. Biswas, A secure light weight scheme for user authentication and key agreement in multi-gateway based wireless sensor networks, Ad. Hoc Netw. 36 (2016) 58–80. https://doi.org/10.1016/j.adhoc.2015.05.020;http://www.sciencedirect.com/science/article/pii/S1570870515001274

[24] R. Amin, S.H. Islam, G.P. Biswas, M.K. Khan, L. Leng, N. Kumar, Design of an anonymity-preserving three-factor authenticated key exchange protocol for wireless sensor networks, Comput. Netw. 101 (C) (2016) 42–62.

[25] S. Arasteh, S.F. Aghili, H. Mala, A new lightweight authentication and key agreement protocol for internet of things, in: Proceedings of the 13th International Iranian Society of Cryptology Conference on Information Security and Cryptology (ISCISC), 2016, pp. 52–59. https://10.1109/ISCISC.2016.7736451

[26] P. Gope, T. Hwang, A realistic lightweight anonymous authentication protocol for securing real-time application data access in wireless sensor networks, IEEE Trans. Ind. Electron. 63 (11) (2016) 7124–7132, doi:10.1109/TIE.2016.2585081.

[27] A. Adavoudi-Jolfaei, M. Ashouri-Talouki, S.F. Aghili, Lightweight and anonymous three-factor authentication and access control scheme for real-time applications in wireless sensor networks, Peer-Peer Netw. Appl. (2017). https://10.1007/s12083-017-0627-8;https://doi.org/10.1007/s12083-017-0627-8

[28] X. Li, M.H. Ibrahim, S. Kumari, A.K. Sangaiah, V. Gupta, K.-K.R. Choo, Anonymous mutual authentication and key agreement scheme for wearable sensors in wireless body area networks, Comput. Netw. 129 (P2) (2017) 429–443. https://10.1016/j.comnet.2017.03.013;https://doi.org/10.1016/j.comnet.2017.03.013

[29] A. Koya, D. P. P., Anonymous hybrid mutual authentication and key agreement scheme for wireless body area network, Comput. Netw. 140 (2018) 138–151.

[30] F. Wu, X. Li, L. Xu, S. Kumari, M. Karuppiah, J. Shen, A lightweight and privacy-preserving mutual authentication scheme for wearable devices assisted by cloud server, Comput. Electr. Eng. 63 (C) (2017) 168–181. https://10.1016/j.compeleceng.2017.04.012;https://doi.org/10.1016/j.compeleceng.2017.04.012

[31] A.K. Das, M. Wazid, N. Kumar, M.K. Khan, K.R. Choo, Y. Park, Design of secure and lightweight authentication protocol for wearable devices environment, IEEE J. Biomed. Health Inf. (2017). https://10.1109/JBHI.2017.2753464

[32] D. Dolev, A.C. Yao, On the security of public key protocols, in: Proceedings of the 22Nd Annual Symposium on Foundations of Computer Science, in: SFCS '81, IEEE Computer Society, Washington, DC, USA, 1981, pp. 350–357, doi:10.1109/SFCS.1981.32.

[33] P. Kocher, J. Jaffe, B. Jun, Differential power analysis, in: M. Wiener (Ed.), Advances in Cryptology — CRYPTO' 99, Springer Berlin Heidelberg, Berlin, Heidelberg, 1999, pp. 388–397.

[34] M. Burrows, M. Abadi, R. Needham, A logic of authentication, ACM Trans. Comput. Syst. 8 (1) (1990) 18–36, doi:10.1145/77648.77649.

[35] AVISPA, AVISPA. Automated Validation of Internet Security Protocols and Applications, May, 2018. http://www.avispa-project.org.

[36] A. Armando, D. Basin, Y. Boichut, Y. Chevalier, L. Compagna, J. Cuellar, P.H. Drielsma, P.-C. Heám, J. Mantovani, S. Mödersheim, D. von Oheimb, M. Rusinowitch, J. Santiago, M. Turuani, L. Viganò, L. Vigneron, The AVISPA tool for the automated validation of internet security protocols and applications, in: K. Etessami, S.K. Rajamani (Eds.), Proceedings of the 17th International Conference on Computer Aided Verification (CAV'05), LNCS, 3576, Springer, 2005, pp. 281–285. Available at http://www.avispa-project.org/publications.html

[37] L. Viganó, Automated security protocol analysis with the Avispa tool, Electr. Notes Theor. Comput. Sci. 155 (2006) 61–86, doi:10.1016/j.entcs.2005.11.052. Proceedings of the 21st Annual Conference on Mathematical Foundations of Programming Semantics (MFPS XXI)

[38] D. Basin, S. Mödersheim, L. Viganò, Ofmc: a symbolic model checker for security protocols, Int. J. Inf. Secur. 4 (3) (2005) 181–208, doi:10.1007/s10207-004-0055-7.

[39] M. Turuani, The Cl-Atse protocol analyser, in: F. Pfenning (Ed.), Term Rewriting and Applications, Springer Berlin Heidelberg, Berlin, Heidelberg, 2006, pp. 277–286.

[40] A. Armando, R. Carbone, L. Compagna, Satmc: a sat-based model checker for security protocols, business processes, and security apis, Int. J. Software Tools Technol. Trans. 18 (2) (2016) 187–204, doi:10.1007/s10009-015-0385-y.



Ankur Gupta received his M. Tech. degree in Computer Science and Engineering from the Motilal Nehru National Institute of Technology Allahabad, India, in 2015. He is currently pursuing his Ph.D. degree in Computer Science and Engineering from the Malaviya National Institute of Technology Jaipur, India. His current research interests include lightweight authentication, cryptography, access control systems, network security and wireless sensor networks.



Meenakshi Tripathi is an Associate Professor at Malaviya National Institute of Technology (MNIT), Jaipur (India). She obtained her Ph.D. from MNIT in 2015. Her research areas are Wireless Sensor Networks (WSN), Software Defined Networks (SDN) & Internet of Things (IoT). She has published more than 30 research papers in refereed International & National Journals & proceedings of National & International conferences. She has presented her research ideas by participating in various conferences within India & outside India like China, Canada, USA etc. She is working on various research projects funded by the Department of Science and Technology, Ministry of Electronics and Information Technology, India of worth around Rs. 50 Lakhs. She is the reviewer for several renowned journals such as International Journal of Communication System (Wiley), Security and Communication Networks (Hindawi), Wireless Communications, etc. She has also been part of several International conferences such as SPACE 2015, ICISS 2016, ICSP 2019 etc. She is an upright member of institute of Electrical & Electronics Engineers (IEEE) & Association of Computing Machinery (ACM). She is also the Chairman of Computer Society of India (Jaipur Chapter).



Tabish Shaikh is currently pursuing his undergraduate degree in Computer Science and Engineering from the Indian Institute of Technology, Jammu. His research interests include cryptography, network security, and machine intelligence.



Aakar Sharma is currently doing his B. Tech. in Computer Science and Engineering from the Indian Institute of Technology Jammu. His current research interests include cryptography, network security, machine learning and deep learning.